

GYM Membership Management System

A MINI PROJECT REPORT

CS2307 – INTERNET PROGRAMMING LAB

Submitted by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2026 – 2027 (ODD SEMESTER)

CS2307 – INTERNET PROGRAMMING LAB

MINI PROJECT

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Title

GYM Membership Management System — A Dynamic Web Application using HTML, CSS, JavaScript, PHP and MySQL.

Abstract

Manual, register-based tracking of gym memberships is slow, error-prone, and gives owners no real-time picture of who is active, who has expired, or how much revenue has been collected. This project, FitZone Gym – Membership Management System, is a dynamic, database-driven website that automates this process end to end. The front end is built with HTML5 and CSS3 for structure and styling and JavaScript for client-side interactivity such as form validation and live UI feedback. The back end is implemented in PHP, which processes form submissions, manages user sessions, and performs full CRUD (Create, Read, Update, Delete) operations against a MySQL relational database. The system supports two roles — Admin and Member — each with a dedicated dashboard, and demonstrates core dynamic-web concepts including secure password hashing, session-based authentication, prepared statements to prevent SQL injection, and server-computed business logic such as automatic membership-expiry calculation.

Introduction

Fitness centers and gyms need an efficient way to manage member registrations, subscription plans, renewals, and payments. Traditional paper registers or standalone spreadsheets do not scale, are hard to search, and cannot notify staff when a membership is about to expire. A web-based Membership Management System solves this by centralizing all member data in a MySQL database that can be accessed securely from any browser through a PHP-powered website.

This mini project, developed for the CS2307 – Internet Programming Lab, showcases the complete internet-programming stack taught in the course: static page structuring with HTML, visual design with CSS, client-side behaviour with JavaScript, and dynamic, data-driven page generation with PHP and MySQL. The chosen domain — a Gym Membership Management System — was selected because it naturally requires several classic dynamic-web features: user registration, authentication, role-based dashboards, and database-backed CRUD operations, making it an ideal demonstration of full-stack web development fundamentals.

Objectives of the Project

- To design a responsive, multi-page website using HTML5 and CSS3 for a fitness/gym business.

- To implement client-side interactivity and form validation using JavaScript.
- To build a secure user registration and login system using PHP sessions and hashed passwords (bcrypt via `password_hash()`).
- To design a normalized MySQL database schema for users, membership plans, members, and payments.
- To implement full CRUD functionality for gym membership records, restricted to an authenticated Admin role.
- To dynamically render content (membership plans, member lists, dashboard statistics) directly from the database rather than hard-coding it.
- To automatically compute and update membership status (active/expired) based on the current date and the plan's duration.
- To deploy the static front-end portion of the project to a live hosting platform (Netlify) as a public demo link.

Modules Description

The system is organized into the following functional modules:

- Public Website Module — Home, About and Contact pages (HTML/CSS/JS) introducing the gym and its facilities, with a client-side validated contact form.
- Membership Plans Module — a PHP page that reads the plans table from MySQL and dynamically renders each plan (Basic, Standard, Premium, Annual) with price and duration.
- Registration Module — allows a visitor to create an account, choose a membership plan, and automatically creates linked rows in the users and members tables, computing the expiry date from the plan duration.
- Authentication Module — a login page that verifies credentials with `password_verify()` and starts a PHP session, redirecting Admin and Member roles to their respective dashboards.
- Admin Dashboard Module — shows live statistics (total, active, expired members, and total revenue), a searchable member table, and links to Add / Edit / Delete member records (full CRUD, restricted to admin role).
- Member Dashboard Module — lets a logged-in member view their own plan, join date, expiry date, and status, with an on-screen warning when the membership is close to expiry.
- Database Module — a MySQL schema (gym_db) with four related tables: users, plans, members, and payments, connected via foreign keys.

Methodology / Approach

The project was developed using the classic LAMP-style workflow (Linux, Apache/PHP built-in server, MySQL, PHP), following these steps:

1. Requirement analysis — identified the core dynamic feature required (registration + login + database-backed membership CRUD).
2. Database design — designed a normalized schema with four tables (users, plans, members, payments) linked by foreign keys, and wrote `gym_db.sql` to create and seed them.
3. UI design — built the static pages (`index.html`, `about.html`, `contact.html`) and a shared stylesheet (`style.css`) using a consistent color theme and a mobile-responsive layout.
4. Client-side scripting — added `script.js` for form validation (login, contact form), dynamic footer year, and simple UX feedback.

- 5. Server-side scripting — implemented PHP scripts using the mysqli extension with prepared statements for every database write, and PHP sessions for authentication and role-based access control.
- 6. Testing — the complete application was run locally on a PHP built-in server connected to a live MySQL database; every dynamic feature (registration, login, plan listing, add/edit/delete member, auto-expiry) was tested end-to-end and the resulting live pages were captured as screenshots (included in this report).
- 7. Deployment — the static portion of the site was published to Netlify as a live demo link, and the complete source code was packaged into a ZIP file for local deployment on any PHP + MySQL server (XAMPP/WAMP/LAMP).

Implementation Code

Below are key excerpts from the project's source code. The complete, runnable source code for every file is provided in the accompanying ZIP file.

1. sql/gym_db.sql — Database schema (excerpt)

```
-- =====
-- GYM MEMBERSHIP MANAGEMENT SYSTEM - DATABASE SCHEMA
-- Database: gym_db
-- =====

CREATE DATABASE IF NOT EXISTS gym_db;
USE gym_db;

-----
-- Table: users (login accounts - admin & members)
-----
CREATE TABLE IF NOT EXISTS users (
  user_id INT AUTO_INCREMENT PRIMARY KEY,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(100) NOT NULL UNIQUE,
  password VARCHAR(255) NOT NULL,
  role ENUM('admin','member') NOT NULL DEFAULT 'member',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-----
-- Table: plans (membership plans)
-----
CREATE TABLE IF NOT EXISTS plans (
  plan_id INT AUTO_INCREMENT PRIMARY KEY,
  plan_name VARCHAR(50) NOT NULL,
  duration_months INT NOT NULL,
  price DECIMAL(10,2) NOT NULL,
  description VARCHAR(255)
);

-----
-- Table: members (gym membership records, linked to users)
```

2. php/config.php — Database connection

```
<?php
/* =====
config.php - Database connection settings
GYM MEMBERSHIP MANAGEMENT SYSTEM
```

```

===== */

session_start();

define('DB_HOST', 'localhost');
define('DB_USER', 'gymuser');
define('DB_PASS', 'gympass123');
define('DB_NAME', 'gym_db');

// Create connection using MySQLi
$conn = new mysqli(DB_HOST, DB_USER, DB_PASS, DB_NAME);

// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
?>

```

3. *php/login.php — Authentication logic (PHP excerpt)*

```

<?php
include 'config.php';

$error = "";

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $email = trim($_POST['email']);
    $password = $_POST['password'];

    $stmt = $conn->prepare("SELECT user_id, full_name, password, role FROM users WHERE email = ?");
    $stmt->bind_param("s", $email);
    $stmt->execute();
    $result = $stmt->get_result();

    if ($result->num_rows === 1) {
        $user = $result->fetch_assoc();
        if (password_verify($password, $user['password'])) {
            $_SESSION['user_id'] = $user['user_id'];
            $_SESSION['full_name'] = $user['full_name'];
            $_SESSION['role'] = $user['role'];

            if ($user['role'] === 'admin') {
                header("Location: admin_dashboard.php");
            } else {
                header("Location: member_dashboard.php");
            }
            exit();
        } else {
            $error = "Incorrect password.";
        }
    } else {

```

4. *php/register.php — Registration logic (PHP excerpt)*

```

<?php
include 'config.php';

$error = "";
$success = "";

```

```

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $full_name = trim($_POST['full_name']);
    $email     = trim($_POST['email']);
    $password  = $_POST['password'];
    $phone     = trim($_POST['phone']);
    $plan_id   = intval($_POST['plan_id']);

    if ($full_name === "" || $email === "" || $password === "") {
        $error = "Please fill all required fields.";
    } else {
        // Check if email already exists
        $stmt = $conn->prepare("SELECT user_id FROM users WHERE email = ?");
        $stmt->bind_param("s", $email);
        $stmt->execute();
        $stmt->store_result();

        if ($stmt->num_rows > 0) {
            $error = "An account with this email already exists.";
        } else {
            $hashed_password = password_hash($password, PASSWORD_BCRYPT);

            // Insert into users table
            $stmt2 = $conn->prepare("INSERT INTO users (full_name, email, password, role) VALUES (?, ?, ?, 'member')");
            $stmt2->bind_param("sss", $full_name, $email, $hashed_password);

            if ($stmt2->execute()) {
                $user_id = $stmt2->insert_id;
                $join_date = date("Y-m-d");

                // Get plan duration to compute expiry date
                $plan_stmt = $conn->prepare("SELECT duration_months FROM plans WHERE plan_id = ?");
                $plan_stmt->bind_param("i", $plan_id);
                $plan_stmt->execute();
                $plan_result = $plan_stmt->get_result();
                $plan_row = $plan_result->fetch_assoc();
                $months = $plan_row ? intval($plan_row['duration_months']) : 1;
                $expiry_date = date("Y-m-d", strtotime("+ $months months"));

                // Insert into members table
                $stmt3 = $conn->prepare("INSERT INTO members (user_id, plan_id, phone, join_date, expiry_date, status) VALUES (?, ?, ?, ?, 'active')");
                $stmt3->bind_param("iisss", $user_id, $plan_id, $phone, $join_date, $expiry_date);
                $stmt3->execute();
            }
        }
    }
}

```

5. [php/admin_dashboard.php](#) — Admin statistics & member listing (PHP excerpt)

```

<?php
include 'config.php';

if (!isset($_SESSION['user_id']) || $_SESSION['role'] !== 'admin') {
    header("Location: login.php");
    exit();
}

// Auto-update expired memberships
$conn->query("UPDATE members SET status='expired' WHERE expiry_date < CURDATE() AND status='active'");

// Stats
$totalMembers = $conn->query("SELECT COUNT(*) AS c FROM members")->fetch_assoc()['c'];

```

```

$activeMembers = $conn->query("SELECT COUNT(*) AS c FROM members WHERE status='active'")->fetch_assoc()['c'];
$expiredMembers = $conn->query("SELECT COUNT(*) AS c FROM members WHERE status='expired'")->fetch_assoc()['c'];
$revenue = $conn->query("SELECT SUM(p.price) AS r FROM members m JOIN plans p ON m.plan_id=p.plan_id")->fetch_assoc()['r'];

// Search / filter
$search = isset($_GET['search']) ? trim($_GET['search']) : "";
$query = "SELECT m.member_id, u.full_name, u.email, m.phone, p.plan_name, m.join_date, m.expiry_date, m.status
        FROM members m
        JOIN users u ON m.user_id = u.user_id
        JOIN plans p ON m.plan_id = p.plan_id";
if ($search !== "") {
    $searchEsc = $conn->real_escape_string($search);
    $query .= " WHERE u.full_name LIKE '%$searchEsc%' OR u.email LIKE '%$searchEsc%'";
}
$query .= " ORDER BY m.member_id DESC";
$members = $conn->query($query);
?>

```

6. js/script.js — Client-side form validation (excerpt)

```

/* =====
FitZone Gym - Membership Management System
Client-side JavaScript
===== */

// Auto-update footer year on every page
document.addEventListener("DOMContentLoaded", function () {
    var yearSpan = document.getElementById("year");
    if (yearSpan) {
        yearSpan.textContent = new Date().getFullYear();
    }
});

// Simple client-side validation for the login form (login.php)
function validateLogin() {
    var email = document.getElementById("email");
    var password = document.getElementById("password");

    if (email && password) {
        if (email.value.trim() === "" || password.value.trim() === "") {
            alert("Please fill in both email and password.");
            return false;
        }
    }
    return true;
}

// Client-side validation + fake submit handling for the contact form (contact.html)
function validateContactForm() {
    var name = document.getElementById("cname").value.trim();

```

7. css/style.css — Styling (excerpt)

```

/* =====
FitZone Gym - Membership Management System
Main Stylesheet
===== */

:root {
    --primary: #ff5722;
    --primary-dark: #e64a19;

```

```
--dark: #1c1c1e;
--light: #f7f7f9;
--gray: #6b6b6f;
--success: #2e7d32;
--error: #c62828;
--radius: 10px;
}

* { margin: 0; padding: 0; box-sizing: border-box; }

body {
  font-family: 'Segoe UI', Arial, sans-serif;
  background: var(--light);
  color: var(--dark);
  line-height: 1.6;
}

a { text-decoration: none; color: inherit; }
ul { list-style: none; }
```

Screenshots

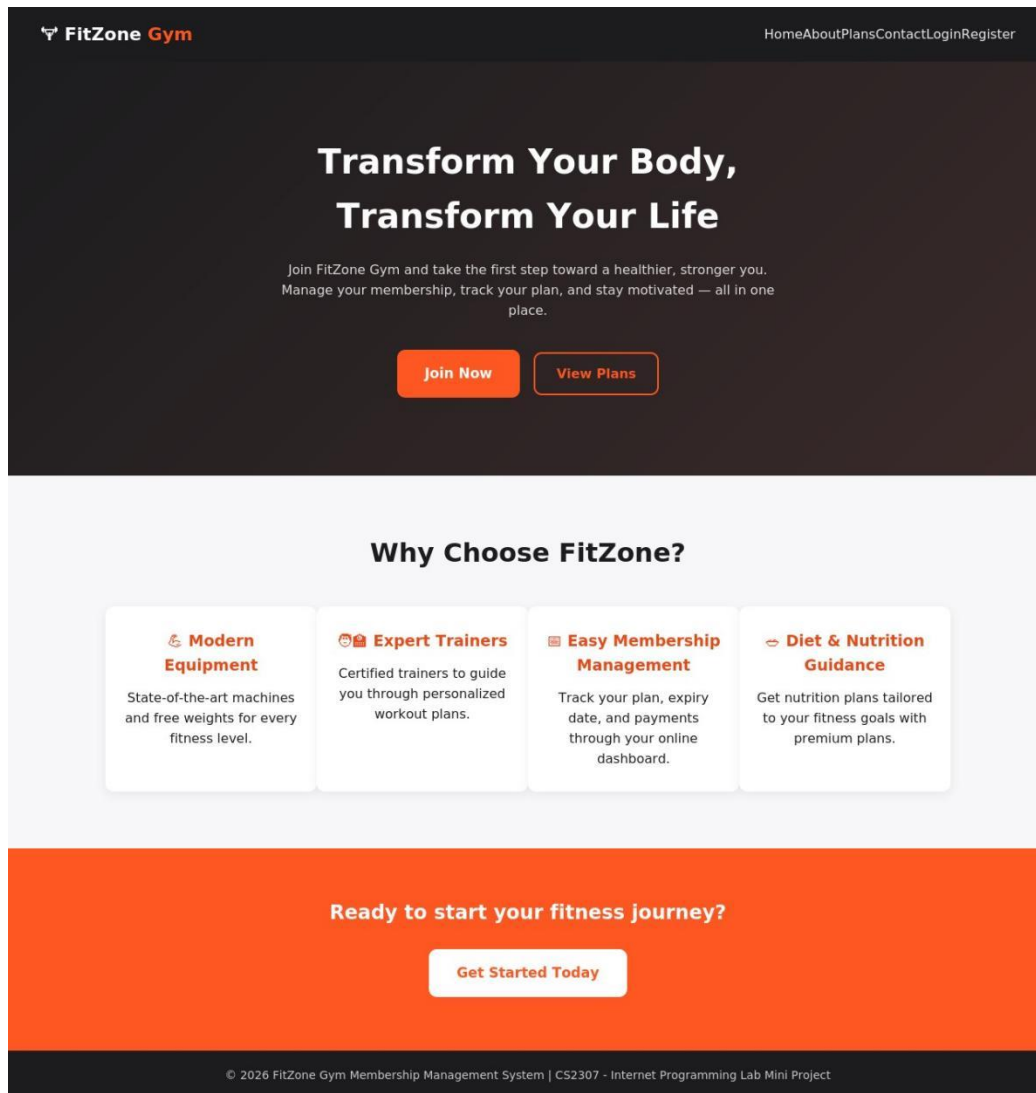


Fig 1. Home page

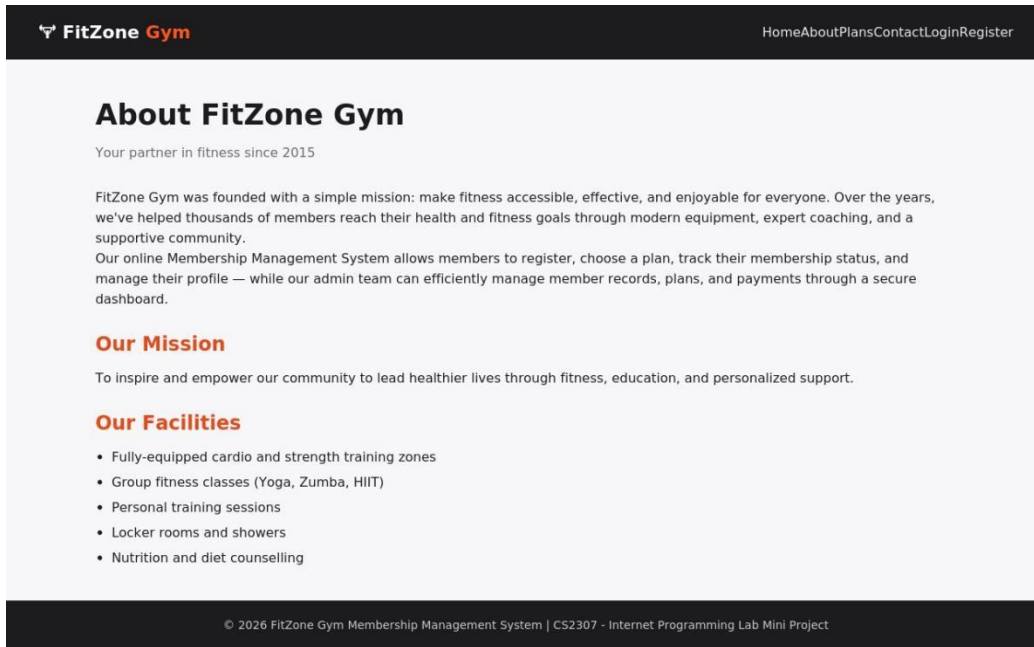


Fig 2. About page

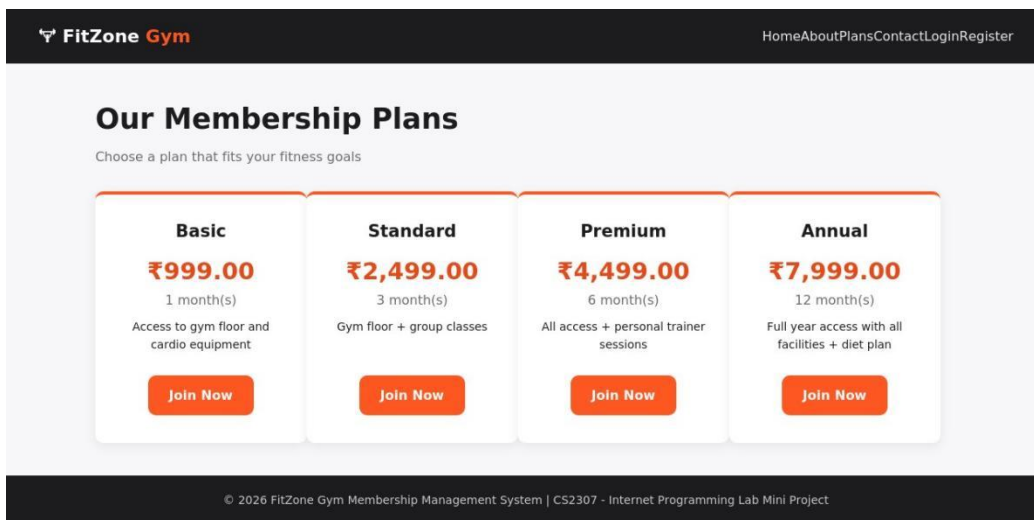


Fig 3. Membership Plans page (rendered dynamically from MySQL)

FitZone Gym

HomeAboutPlansContactLoginRegister

Create Your Account

Full Name

Email

Password

Phone

Membership Plan

Basic - ₹999.00 (1 months) ▼

Register

Already have an account? [Login](#)

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Fig 4. Registration page

FitZone Gym

HomeAboutPlansContactLoginRegister

Member / Admin Login

Email

Password

Login

Don't have an account? [Register](#)

Demo admin login: admin@gym.com / admin123

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Fig 5. Login page

[Home](#)
[About](#)
[Plans](#)
[Contact](#)
[Login](#)
[Register](#)

Get In Touch

Have a question? Send us a message.

Name

Email

Message

Send Message

📍 123 Fitness Street, Chennai, Tamil Nadu

📞 +91 98765 43210

✉ info@fitzonegym.com

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Fig 6. Contact page with client-side validated form

[Home](#)
[About](#)
[Plans](#)
[Contact](#)
[Dashboard](#)
[Logout \(Gym Admin\)](#)

Admin Dashboard

3

Total Members

2

Active

1

Expired

₹13,497.00

Total Revenue

ID	Name	Email	Phone	Plan	Join Date	Expiry Date	Status	Actions
5	Divya Raj	divya@example.com	9012345678	Annual	2026-09-25	2027-09-25	Active	Edit Delete
4	Arun Kumar	arun@example.com	9988776655	Basic	2026-06-01	2026-07-01	Expired	Edit Delete
3	Priya Sharma	priya@example.com	9123456780	Premium	2026-09-25	2027-03-25	Active	Edit Delete

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Fig 7. Admin Dashboard — live statistics and member CRUD table

The screenshot shows the 'Add New Member' form in the FitZone Gym Admin interface. The form is centered on a light gray background. It includes input fields for 'Full Name', 'Email', 'Phone', and 'Join Date'. A dropdown menu for 'Plan' is set to 'Basic'. Below the form is an orange 'Add Member' button and a link to 'Back to Dashboard'. The header shows the FitZone Gym logo and navigation links: Home, About, Plans, Contact, Dashboard, and Logout (Gym Admin). The footer contains the copyright notice: © 2026 FitZone Gym Membership Management System | CS2307 - Internet Programming Lab Mini Project.

FitZone Gym HomeAboutPlansContactDashboardLogout (Gym Admin)

Add New Member

Full Name

Email

Phone

Plan
Basic

Join Date
2026-09-25

Add Member

[← Back to Dashboard](#)

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Fig 8. Admin — Add New Member form

The screenshot shows the Member Dashboard in the FitZone Gym self-service membership view. The dashboard is centered on a light gray background. It features a 'Welcome, Priya Sharma' message with a hand icon. Below this is a white box containing member details: Email (priya@example.com), Phone (9123456780), Current Plan (Premium (₹4,499.00)), Join Date (2026-09-25), Expiry Date (2027-03-25), and Status (Active). The header shows the FitZone Gym logo and navigation links: Home, About, Plans, Contact, My Account, and Logout (Priya Sharma). The footer contains the copyright notice: © 2026 FitZone Gym Membership Management System | CS2307 - Internet Programming Lab Mini Project.

FitZone Gym HomeAboutPlansContactMy AccountLogout (Priya Sharma)

Welcome, Priya Sharma

Email: priya@example.com
Phone: 9123456780
Current Plan: Premium (₹4,499.00)
Join Date: 2026-09-25
Expiry Date: 2027-03-25
Status: Active

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Fig 9. Member Dashboard — self-service membership view

References

- PHP Official Documentation — <https://www.php.net/docs.php>
- MySQL 8.0 Reference Manual — <https://dev.mysql.com/doc/>
- MDN Web Docs (HTML, CSS, JavaScript) — <https://developer.mozilla.org/>
- W3Schools PHP & MySQL Tutorials — <https://www.w3schools.com/php/>
- Netlify Documentation — <https://docs.netlify.com/>
- CS2307 — Internet Programming Lab, Course Manual (Department of Computer Science and Engineering).

Conclusion

The GYM Membership Management System successfully demonstrates a complete dynamic web application built with HTML, CSS, JavaScript, PHP, and MySQL. It covers the full spectrum of concepts taught in the Internet Programming Lab: static page design, client-side scripting, server-side form processing, session-based authentication, and relational database connectivity with full CRUD operations. The system automates a real-world workflow — gym membership registration, tracking, and renewal — replacing manual record-keeping with a searchable, role-based, always-up-to-date dashboard. All dynamic features (registration, login, plan browsing, member management, and auto-expiry) were tested end-to-end against a live MySQL database, and the working application was captured in the screenshots above. The project can be extended in the future with online payment gateway integration, SMS/email renewal reminders, attendance tracking via QR/biometric check-in, and a REST API for a companion mobile app.